Part 1 — New Features & Data Design

The system contain 4 main entities

- Accounts
- categories
- Transactions
- Transfers

Calculated values (not stored)

- Net worth
- Monthly summaries
- Dashboard metrics

Categories

Categories

id: int PK

name: string

type: string

Default categories

- Salary (income)
- Freelance (income)
- Rent (expense)
- Groceries (expense)

Business Rules

- Each transaction must belong to one category
- User can add/delete thier own categories
- System seeds default categories when database is empty

Dates on Transaction

transactions

id: int PK

amount: decimal

date: date

account_id: int FK

category_id: int FK

Business Rules

- Every transaction must have a date
- Transaction can be filtered by month and year

Multiple Accounts

Accounts

id: int PK

name: string

opening balance: decimal

Business Rules

- users can now manage multiple accounts
- Account balances can be negative
- credit card accounts starts at 0 and becomes negative when spending

current balance = opening balance + transaction(income) - transaction(expense)

Transfer between accounts

Transfer

id: int PK

amount: decimal

date: date

description: string

from_account_id : int FK

to_account_id : int FK

Business Rules

- user transfer money from one account to another
- A transfer must have source account
- A transfer must have destination account

Relationships

- Category → Transactions: one to many

  one category can contain many transactions, but each transaction belongs to only one category

- Account → Transactions: one to many

  one account can have many transactions, but each transaction belongs to only one account

- Account → Transfers (source): one to many

  one account can send money in many transfers as the source account

- Account → Transfers (destination): one to many

  one account can receive money in many transfers as the destination account

Net Worth

Business Rules

- The application must be able to show the user their net show
- read only value – its never stored
- calculated dynamically

Net worth = Sum of all current account balances

Import / Export (Data Portability)

Business Rules

- export all data to a .zip file and importing it back

if validation fails	if validation succeeds:
show a clear error message	Delete current data
Do not modify existing data	Replace with imported data